Aby Mathew

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Systems & Local-First Software Developer `summer/fall '26`

Undergraduate systems builder focused on resilient offline-first software, hardened runtimes, and low-latency local compute. Track record of reverse-engineering legacy systems, eliminating cloud dependencies, and building rock-solid C#, Node.js, and C applications that do not break under network failure.

TECHNICAL SKILLS & WORKBENCH

Languages: C, Python, C# (.NET 8), JavaScript / TypeScript, SQL, Bash
Systems & Runtime: Linux, POSIX Sockets, Win32 P/Invoke, SQLite (WAL mode), Docker, Cloudflare Workers
Core Competencies: Local-First Architecture, TCP Socket Management, Anti-SSRF / Network Security, RAG Pipelines

SELECTED ENGINEERING PROJECTS

Johns Fitness — Edge & Access Control System

Lead Architect & Developer | 2026 – Present | github.com/4ofl/johnsfitness-edge ↗

Offline-first biometric access control daemon & reception PWA for 1,405 active gym members.

- Engineered .NET 8 x86 background service interfacing with BioMax optical & facial DSP terminals via Win32 P/Invoke over LAN TCP socket pool.
- Decoupled door relay from host OS and cloud: on-device NVRAM DSP matching trips 5,000ms pulse in < 15ms autonomous fast-path.
- Designed transactional sync outbox backed by local SQLite WAL mode, guaranteeing 100% check-in continuity during ISP outages.
- Reverse-engineered and migrated legacy 2013 Microsoft Access (.mdb) database containing 1,405 member records to SQLite with zero data corruption.

June — Hardened Discord Application & Infrastructure

Author & Maintainer | 2026 – Present | docs.june.4ofl.dev ↗

High-reliability Discord application with zero C++ database dependencies.

- Built on Node.js 22 LTS utilizing built-in `node:sqlite` in WAL mode, eliminating native C++ toolchain dependencies.
- Mitigated SSRF and DNS rebinding (TOCTOU) attacks at raw socket level by validating and pinning remote IP addresses prior to TCP handshake.
- Engineered custom gateway heartbeat state machine recovering from silent half-open TCP connections dropped without close frames.
- Isolated user-supplied regex execution in isolated Worker Thread watchdogs with 500ms hard deadlines to prevent ReDoS CPU starvation.

Citecamp — Academic RAG Document Pipeline

Developer | 2026 | 4ofl.dev/work/citecamp/ ↗

Specialized RAG pipeline for scanned, watermarked institutional circulars.

- Engineered pre-retrieval OCR cleaning and document layout pipeline tailored for low-quality scanned university PDFs.
- Maintained deterministic institutional source citation hashes attached to every LLM retrieval step.

EDUCATION & CREDENTIALS

Bachelor of Computer Applications (BCA)

Undergrad College / University | 2026 – Present | Kerala, India

Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks.